how to learn math for machine learning fast (even with zero math background)_transcript_20260201_132646

Source: How To Learn Math for Machine Learning FAST (Even With Zero Math Background).mp3
Language Mode: Auto-detect (Mixed)
Detected: en (99.9%)
Transcribed: 2026-02-01 13:28:55

==========

Do you want to become a data scientist or machine learning engineer, but you feel intimidated by all ; the math involved? I get it, I've been there. I dropped out of high school after 10th grade, ; so I never learned any math beyond trigonometry in school. When I started my journey into machine ; learning, I didn't even know what a derivative was. Fast forward to today, and I'm an applied ; scientist at Amazon, and I feel pretty confident in my math skills. I've picked up the necessary ; math along the way using free resources and self-directed learning. Today I'm going to walk ; you through some of my favorite books, courses, and YouTube channels that helped me to get where ; I am today, and I'll also share some study tips on how to study effectively and not waste your ; time struggling and being bored. So let's get to it. First, let's address a common question. Do you ; even really need to know the math to work in machine learning? The short answer is, it depends ; on what you want to do. For research-heavy roles where you're creating new machine learning ; algorithms, then yes, you obviously need to know the math. But if you're asking yourself if you ; need to learn math, chances are that's not the kind of job you're looking for. For practitioners, ; most of us in the industry, you can be totally competent without knowing all the underlying ; details. ; At this point, libraries like NumPy, Scikit Learn, and TensorFlow handle most of the heavy ; lifting for you. You don't need to know the math behind gradient descent to deploy a model to ; production. If you're a beginner trying to get into machine learning, in my opinion it's not ; strategic to spend a bunch of time memorizing formulas or studying linear algebra. You should ; be spending that time building things. Train a simple model, explore your data, build a pipeline ; that predicts something fun. That said, there are moments where knowing the math really helps. ; Here are a couple of examples. So imagine you're training a model, and it's just not converging. ; If you understand concepts like gradients and optimization functions, you'll know whether to ; adjust your learning rate, try a different optimizer, or tweak your data processing. ; Or let's say you're running a linear regression, and you're interpreting the coefficients. Without ; math knowledge, you might miss problems like multicolinearity, which makes the coefficients ; unreliable. Then you might make incorrect conclusions from the data, and cost the company ; millions and lose your job. Just kidding. Kind of. We do really need to be careful when making ; business decisions from the models that we build. So while you can, and should, get started without ; deep math knowledge, you should be careful when making business decisions from the models that we ; build. It's definitely reasonable to build your comfort with math over time. Once you're

hands-on, ; you'll start encountering problems that naturally push you to learn more. When you need to debug or ; explain your results, that's when the math is going to start to click because it's connected ; to something real. So seriously, don't let fear of math stop you from starting. You don't need to ; learn it all up front to make progress. Get your hands dirty with the tools, build your portfolio, ; and let math grow as a skill alongside your practical knowledge. Alright, now let's talk ; about what to learn when you're building your math foundation for machine learning jobs. ; First, linear algebra. ; Linear algebra is fundamental for machine learning, especially for deep learning. Many ; models rely on representing data and computations as matrices and vectors. Here's what to prioritize. ; First, matrices and vectors. Think of matrices as grids of numbers and vectors as lists of numbers. ; Data is often stored this way, and operations like addition, multiplication, and dot products ; are really central to how models process that information. Determinants and inverses. ; Determinants tell you whether a matrix can be inverted, which is used in optimization problems ; and solving systems of linear equations. Eigenvalues and eigenvectors. ; These are key to understanding variance in data and are foundational of techniques like ; principal component analysis, which can help to reduce the dimensionality in data sets. ; Lastly, matrix decomposition. Methods like singular value decomposition are used in recommendation ; systems, dimensionality reduction, and data compression. ; Now we're on to basic calculus. Calculus is core to understanding how models learn ; from data. But we don't need to worry about solving super complex problems. It's just ; about grasping a couple of key ideas. ; First, derivatives and gradients. Derivatives measure how things change, and gradients, ; which are multidimensional derivatives, are what we call gradients. ; power optimization algorithms like gradient descent. These help models adjust their parameters ; to minimize error. The chain rule is central to neural networks. It's how backpropagation works, ; which is the process of figuring out how much each weight in the network contributes to the ; overall error so the model can learn effectively. Lastly, optimization basics. Concepts like local ; versus global minima, saddle points, and convexity are important to understand why some models get ; stuck and others manage to find the best solutions. Lastly, statistics and probability. ; Statistics and probability are the bread and butter of understanding data. While they're ; kind of more associated with data science, there's definitely a lot of value for machine learning as ; well. Here's what you need to know. Distributions. Get familiar with common distributions like the ; normal, binomial, and uniform. The normal distribution in particular pops up everywhere ; in data science and machine learning. Variance and covariance. Variance tells you how spread out ; your data is, while covariance shows you how two variables relate. These concepts are really ; important for feature selection and understanding your data structure. Bayes' theorem. While it has ; kind of a silly, maybe intimidating name, Bayes' theorem is a pretty simple but powerful tool for ; probabilistic reasoning. It's foundational for algorithms like naive Bayes, big surprise, ; which is used for things like spam detection, as well as for Bayesian optimization for hyperparameter ; tuning. You'll also want to understand maximum likelihood estimation, which helps to estimate ; model parameters by finding values that maximize the likelihood of your data. It's a really ; fundamental concept, especially in algorithms like logistic regression. Finally, sampling and ; conditional probability. Sampling lets you work with subsets of data efficiently, and conditional ; probability is essential for understanding the relationship between events, especially in Bayesian models. ; This is definitely not an exhaustive list, but I think it's a good overview of the common concepts ;

you'll need to know to do a good job as a data scientist who's focused on machine learning or as ; a machine learning engineer. Next up, I'll share the best resources to learn these concepts without ; it being stressful or overwhelming, and hopefully also not too boring. Personally, I would highly ; recommend starting with a visual and intuitive understanding of the key concepts before you start ; reading really difficult textbooks and trying to solve equations. For linear algebra and calculus, ; I cannot speak highly enough about 3Blue1Brown's Essence of Linear Algebra and Essence of Calculus series. ; These videos give a really solid introduction to what is actually being measured and manipulated ; when we use these mathematical approaches. More importantly, they kind of show the beauty in it. ; It's kind of strange to say that math videos could be inspirational in a way, but these ones are. ; For statistics and probability, I am a huge fan of StatQuest. His videos are clear, ; engaging, and just a joy to watch. StatQuest has playlists with overviews on core stats ; and machine learning concepts, so I'd recommend starting there. Once you have a visual intuition, ; you can start working through more structured books and courses. There are a lot of great ; options here as well. Let's go through a few that I personally use to learn. I completed the ; Mathematics for Machine Learning specialization from Imperial College London on Coursera when I ; was just starting out. The specialization is divided into three courses, linear algebra, ; multivariate calculus, and a last one on principal component analysis. The courses are well structured ; and include a mix of video lectures, quizzes, and programming assignments in Python. I found the ; course to be a little bit challenging as a beginner, but it was a really good overview and I ; did manage to pass everything with a little bit of effort. DeepLearning.ai also recently released a ; specialization on Coursera. This specialization has courses on linear algebra and calculus, ; but instead of PCA, the final course focuses on stats and probability. I'm personally working ; through this specialization right now and overall I'm finding it to be another really great option. ; Each module starts with a nice motivation for how the math connects to an applied machine learning ; concept. It has coding exercises in Python and some neat 3D tools to mess around with to get a ; good kind of visual understanding of the concepts as well. If you prefer learning from books, ; I have some suggestions there too. First up, if you like anime or nerdy stuff, boy oh boy do I have a ; recommendation for you. Did you know that they have manga math books? So I've read the Manga ; Guide to Calculus, Manga Guide to Linear Algebra, and Manga Guide to Statistics. They also have a ; ton more. They have some on like physics, I think, and some other engineering concepts. So they're ; super super fun. I can't say the instructional quality is like world class or anything, ; but they're cute and engaging and they made me not dread reading a math book. ; The next level up would be real math books. These are some of the best. ; The Mathematics for Machine Learning eBook by Dysonroth and colleagues is a great, ; comprehensive resource and it's available for free for personal use. It covers key topics ; we've already discussed like linear algebra, calculus, probability, and optimization, ; with a focus on how these concepts apply to machine learning algorithms. It's relatively ; beginner friendly and is generally regarded as one of the best books for learning this material. ; Next, Practical Statistics for Data Scientists is another well-loved resource that includes code ; examples in Python and R. Now before we actually start studying, I think it's important to spend ; a little bit of time thinking really deeply about why you even want to do this. Personally, I find ; studying just because I feel like I should or because it's some arbitrary assignment. I get ; distracted easily and I don't actually end up retaining that much of the

information. ; Instead, I try to connect to some kind of a deeper motivation, if possible. ; Personally, right now I have a really basic motivation. I want to do well in my job because ; I want to earn a lot of money so that I can take care of everyone I love. I have these great ; resources to learn from that weren't even available a couple of years ago, and I have this opportunity ; to push myself and make sure that everyone I love is safe and cared for now and in the future. ; So this isn't to put extra pressure on myself, but actually just a way that works for me to kind ; of get excited that I have this opportunity to learn and grow and hopefully help others along ; the way. Your motivation might be totally different, but whatever it is, try to tie ; this work to a larger goal. In terms of optimizing your study time, I've found that one of the most ; effective methods is writing notes in my own words. Don't just copy definitions or formulas. ; Take time to summarize concepts as if you were explaining them to someone else, or really to ; future you. For example, if you're learning about derivatives, you might write, ; a derivative measures how a function changes given a tiny nudge to its input. This forces ; you to actively process the material. Relatedly, when it comes to math formulas, don't just stare ; at them. I know it's easy to do. It's really helpful to translate them into plain English or ; whatever spoken language you prefer. For instance, take the equation y equals mx plus b. You might ; describe m as the slope that shows how steep the line is and b as the point where the line crosses ; the y-axis. So the final formula might be the value of y, the output, is determined by taking the slope ; by x, the input, and then adding b, the starting point where the line intersects the y-axis. ; You can even use your notes as like a personal blog, kind of. ; Writing short posts about what you've learned is a really solid way to clarify your understanding, ; and teaching others, even if nobody reads it, solidifies the material in your own mind. ; Plus, sharing your posts on Medium or LinkedIn not only potentially helps others, ; but also allows you to build kind of a portfolio showcasing your learning journey. ; Also, trust me, when it's interview time, you're going to be really happy that you have these ; notes. I use my own study time, so I'm going to use my own study time. I'm going to use my own ; study notes all the time. The next piece of advice I have might not be super duper fun, ; but I also recommend not using just one resource. Personally, I've had a lot of success from taking ; many different courses and kind of just throwing all my notes together at first. Then I'll write ; a blog, like I was just talking about, that summarizes all of my learnings. There are a ; couple of advantages to this approach. First, repetition helps you retain things. If I see ; a concept multiple times, explained from multiple angles, I'm much more likely to actually get ; what's going on and remember it for longer than a day. Plus, not only do I see the information ; presented to me multiple times, I'm writing the concepts out in my own words multiple times, ; including that final time where I synthesize it all and get ready to share with others. ; So I have to be really confident that I actually got it by the end if I'm going to be putting it ; out publicly. Let me show you an example. Here are three different explanations for the concept ; of rank in linear algebra. Starting with three blue, one brown, you can see this visual intuition, ; which is a great starting point, but you know, it lacks a little detail. Then we can look at how ; rank was presented in deeplearning.ai, which has some more detail, but is still presented in a ; really simple way. Finally, we can go to the Mathematics for Machine Learning ebook to get ; an even more formalized definition with additional detail. Each one of these builds on the other and ; adds to my store of knowledge. Finally, once you've built that foundation and get to the level ; of math where you can actually use it for stuff, I really recommend coding concepts from scratch. ; If you can code

gradient descent or logistic regression using just NumPy, you're off to a ; really strong start. While I know at this point you are super excited to start learning math, ; I do want to just circle back to the important fact that if you're a beginner trying to get your ; first job, in my opinion math should not be the first thing you prioritize. It's really unlikely ; that your math skills are going to be the thing that gets you a job as a data scientist or machine ; learning engineer. Instead, prioritize gaining hands-on experience by working on projects and ; actually building stuff. Employers are far more interested in seeing what you can do with the ; tools and knowledge you already have than how many formulas you've memorized. As you encounter ; challenges in your work, you'll naturally be motivated to learn the math behind the algorithms. ; Remember, math is a tool to help you succeed and shouldn't be a barrier to getting started. Now, ; if you're curious about what else you need to know to become a data scientist, ; I have a really comprehensive video that lays out the entire roadmap from absolute beginner ; to your first job, including skills to learn, project examples, how to get experience, ; preparing for interviews, and more. Definitely check that out next!

========== TIMESTAMPS:

[00:00 - 00:04] Do you want to become a data scientist or machine learning engineer, but you feel intimidated by all

[00:04 - 00:08] the math involved? I get it, I've been there. I dropped out of high school after 10th grade,

[00:08 - 00:13] so I never learned any math beyond trigonometry in school. When I started my journey into machine

[00:13 - 00:16] learning, I didn't even know what a derivative was. Fast forward to today, and I'm an applied

[00:16 - 00:20] scientist at Amazon, and I feel pretty confident in my math skills. I've picked up the necessary

[00:20 - 00:25] math along the way using free resources and self-directed learning. Today I'm going to walk

[00:25 - 00:28] you through some of my favorite books, courses, and YouTube channels that helped me to get where

[00:28 - 00:32] I am today, and I'll also share some study tips on how to study effectively and not waste your

[00:32 - 00:38] time struggling and being bored. So let's get to it. First, let's address a common question. Do you

[00:38 - 00:41] even really need to know the math to work in machine learning? The short answer is, it

depends

[00:41 - 00:45] on what you want to do. For research-heavy roles where you're creating new machine learning

[00:45 - 00:49] algorithms, then yes, you obviously need to know the math. But if you're asking yourself if you

[00:49 - 00:53] need to learn math, chances are that's not the kind of job you're looking for. For practitioners,

[00:53 - 00:57] most of us in the industry, you can be totally competent without knowing all the underlying

[00:57 - 00:58] details.

[00:58 - 01:03] At this point, libraries like NumPy, Scikit Learn, and TensorFlow handle most of the heavy

[01:03 - 01:07] lifting for you. You don't need to know the math behind gradient descent to deploy a model to

[01:07 - 01:11] production. If you're a beginner trying to get into machine learning, in my opinion it's not

[01:11 - 01:15] strategic to spend a bunch of time memorizing formulas or studying linear algebra. You should

[01:15 - 01:20] be spending that time building things. Train a simple model, explore your data, build a pipeline

[01:20 - 01:24] that predicts something fun. That said, there are moments where knowing the math really helps.

[01:24 - 01:28] Here are a couple of examples. So imagine you're training a model, and it's just not converging.

[01:28 - 01:32] If you understand concepts like gradients and optimization functions, you'll know whether to

[01:32 - 01:36] adjust your learning rate, try a different optimizer, or tweak your data processing.

[01:36 - 01:40] Or let's say you're running a linear regression, and you're interpreting the coefficients. Without

[01:40 - 01:44] math knowledge, you might miss problems like multicolinearity, which makes the coefficients

[01:44 - 01:48] unreliable. Then you might make incorrect conclusions from the data, and cost the company

[01:48 - 01:52] millions and lose your job. Just kidding. Kind of. We do really need to be careful when making

[01:52 - 01:57] business decisions from the models that we build. So while you can, and should, get started without

[01:57 - 01:58] deep math knowledge, you should be careful when making business decisions from the models that we

[01:58 - 02:01] build. It's definitely reasonable to build your comfort with math over time. Once you're hands-on,

[02:02 - 02:06] you'll start encountering problems that naturally push you to learn more. When you need to debug or

[02:06 - 02:09] explain your results, that's when the math is going to start to click because it's connected

[02:09 - 02:14] to something real. So seriously, don't let fear of math stop you from starting. You don't need to

[02:14 - 02:18] learn it all up front to make progress. Get your hands dirty with the tools, build your portfolio,

[02:18 - 02:23] and let math grow as a skill alongside your practical knowledge. Alright, now let's talk

[02:23 - 02:25] about what to learn when you're building your math foundation for machine learning jobs.

[02:26 - 02:27] First, linear algebra.

[02:27 - 02:31] Linear algebra is fundamental for machine learning, especially for deep learning. Many

[02:31 - 02:35] models rely on representing data and computations as matrices and vectors. Here's what to prioritize.

[02:36 - 02:41] First, matrices and vectors. Think of matrices as grids of numbers and vectors as lists of numbers.

[02:41 - 02:45] Data is often stored this way, and operations like addition, multiplication, and dot products

[02:45 - 02:49] are really central to how models process that information. Determinants and inverses.

[02:49 - 02:53] Determinants tell you whether a matrix can be inverted, which is used in optimization problems

[02:53 - 02:56] and solving systems of linear equations. Eigenvalues and eigenvectors.

[02:57 - 03:00] These are key to understanding variance in data and are foundational of techniques like

[03:00 - 03:04] principal component analysis, which can help to reduce the dimensionality in data sets.

[03:04 - 03:09] Lastly, matrix decomposition. Methods like singular value decomposition are used in recommendation

[03:09 - 03:11] systems, dimensionality reduction, and data compression.

[03:12 - 03:15] Now we're on to basic calculus. Calculus is core to understanding how models learn

[03:15 - 03:19] from data. But we don't need to worry about solving super complex problems. It's just

[03:19 - 03:20] about grasping a couple of key ideas.

[03:21 - 03:25] First, derivatives and gradients. Derivatives measure how things change, and gradients,

[03:25 - 03:27] which are multidimensional derivatives, are what we call gradients.

[03:27 - 03:31] power optimization algorithms like gradient descent. These help models adjust their parameters

[03:31 - 03:35] to minimize error. The chain rule is central to neural networks. It's how backpropagation works,

[03:35 - 03:39] which is the process of figuring out how much each weight in the network contributes to the

[03:39 - 03:44] overall error so the model can learn effectively. Lastly, optimization basics. Concepts like local

[03:44 - 03:48] versus global minima, saddle points, and convexity are important to understand why some models get

[03:48 - 03:52] stuck and others manage to find the best solutions. Lastly, statistics and probability.

[03:52 - 03:56] Statistics and probability are the bread and butter of understanding data. While they're

[03:56 - 03:59] kind of more associated with data science, there's definitely a lot of value for machine learning as

[03:59 - 04:04] well. Here's what you need to know. Distributions. Get familiar with common distributions like the

[04:04 - 04:08] normal, binomial, and uniform. The normal distribution in particular pops up everywhere

[04:08 - 04:12] in data science and machine learning. Variance and covariance. Variance tells you how spread out

[04:12 - 04:16] your data is, while covariance shows you how two variables relate. These concepts are really

[04:16 - 04:21] important for feature selection and understanding your data structure. Bayes' theorem. While it has

[04:21 - 04:25] kind of a silly, maybe intimidating name, Bayes' theorem is a pretty simple but powerful tool for

[04:25 - 04:29] probabilistic reasoning. It's foundational for algorithms like naive Bayes, big surprise,

[04:29 - 04:34] which is used for things like spam detection, as well as for Bayesian optimization for hyperparameter

[04:34 - 04:38] tuning. You'll also want to understand maximum likelihood estimation, which helps to estimate

[04:38 - 04:42] model parameters by finding values that maximize the likelihood of your data. It's a really

[04:42 - 04:46] fundamental concept, especially in algorithms like logistic regression. Finally, sampling and

[04:46 - 04:51] conditional probability. Sampling lets you work with subsets of data efficiently, and conditional

[04:51 - 04:55] probability is essential for understanding the relationship between events, especially in Bayesian models.

[04:55 - 04:59] This is definitely not an exhaustive list, but I think it's a good overview of the common concepts

[04:59 - 05:03] you'll need to know to do a good job as a data scientist who's focused on machine learning or as

[05:03 - 05:07] a machine learning engineer. Next up, I'll share the best resources to learn these concepts without

[05:07 - 05:11] it being stressful or overwhelming, and hopefully also not too boring. Personally, I would highly

[05:11 - 05:16] recommend starting with a visual and intuitive understanding of the key concepts before you start

[05:16 - 05:20] reading really difficult textbooks and trying to solve equations. For linear algebra and calculus,

[05:20 - 05:25] I cannot speak highly enough about 3Blue1Brown's Essence of Linear Algebra and Essence of Calculus series.

[05:25 - 05:29] These videos give a really solid introduction to what is actually being measured and manipulated

[05:29 - 05:33] when we use these mathematical approaches. More importantly, they kind of show the beauty in it.

[05:33 - 05:37] It's kind of strange to say that math videos could be inspirational in a way, but these ones are.

[05:38 - 05:42] For statistics and probability, I am a huge fan of StatQuest. His videos are clear,

[05:42 - 05:47] engaging, and just a joy to watch. StatQuest has playlists with overviews on core stats

[05:47 - 05:51] and machine learning concepts, so I'd recommend starting there. Once you have a visual intuition,

[05:52 - 05:55] you can start working through more structured books and courses. There are a lot of great

[05:55 - 05:59] options here as well. Let's go through a few that I personally use to learn. I completed the

[05:59 - 06:03] Mathematics for Machine Learning specialization from Imperial College London on Coursera when I

[06:03 - 06:07] was just starting out. The specialization is divided into three courses, linear algebra,

[06:07 - 06:12] multivariate calculus, and a last one on principal component analysis. The courses are well structured

[06:12 - 06:17] and include a mix of video lectures, quizzes, and programming assignments in Python. I found the

[06:17 - 06:20] course to be a little bit challenging as a beginner, but it was a really good overview and I

[06:20 - 06:25] did manage to pass everything with a little bit of effort. DeepLearning.ai also recently released a

[06:25 - 06:29] specialization on Coursera. This specialization has courses on linear algebra and calculus,

[06:30 - 06:34] but instead of PCA, the final course focuses on stats and probability. I'm personally working

[06:34 - 06:37] through this specialization right now and overall I'm finding it to be another really great option.

[06:38 - 06:42] Each module starts with a nice motivation for how the math connects to an applied machine learning

[06:42 - 06:46] concept. It has coding exercises in Python and some neat 3D tools to mess around with to get a

[06:46 - 06:50] good kind of visual understanding of the concepts as well. If you prefer learning from books,

[06:50 - 06:55] I have some suggestions there too. First up, if you like anime or nerdy stuff, boy oh boy do I have a

[06:55 - 07:01] recommendation for you. Did you know that they have manga math books? So I've read the Manga

[07:01 - 07:05] Guide to Calculus, Manga Guide to Linear Algebra, and Manga Guide to Statistics. They also have a

[07:05 - 07:09] ton more. They have some on like physics, I think, and some other engineering concepts. So they're

[07:09 - 07:14] super super fun. I can't say the instructional quality is like world class or anything,

[07:14 - 07:18] but they're cute and engaging and they made me not dread reading a math book.

[07:18 - 07:21] The next level up would be real math books. These are some of the best.

[07:22 - 07:25] The Mathematics for Machine Learning eBook by Dysonroth and colleagues is a great,

[07:25 - 07:29] comprehensive resource and it's available for free for personal use. It covers key topics

[07:29 - 07:33] we've already discussed like linear algebra, calculus, probability, and optimization,

[07:33 - 07:38] with a focus on how these concepts apply to machine learning algorithms. It's relatively

[07:38 - 07:41] beginner friendly and is generally regarded as one of the best books for learning this material.

[07:42 - 07:46] Next, Practical Statistics for Data Scientists is another well-loved resource that includes code

[07:46 - 07:51] examples in Python and R. Now before we actually start studying, I think it's important to spend

[07:51 - 07:55] a little bit of time thinking really deeply about why you even want to do this. Personally, I find

[07:55 - 08:00] studying just because I feel like I should or because it's some arbitrary assignment. I get

[08:00 - 08:03] distracted easily and I don't actually end up retaining that much of the information.

[08:03 - 08:07] Instead, I try to connect to some kind of a deeper motivation, if possible.

[08:07 - 08:12] Personally, right now I have a really basic motivation. I want to do well in my job because

[08:12 - 08:15] I want to earn a lot of money so that I can take care of everyone I love. I have these great

[08:15 - 08:20] resources to learn from that weren't even available a couple of years ago, and I have this

opportunity

[08:20 - 08:25] to push myself and make sure that everyone I love is safe and cared for now and in the future.

[08:25 - 08:29] So this isn't to put extra pressure on myself, but actually just a way that works for me to kind

[08:29 - 08:34] of get excited that I have this opportunity to learn and grow and hopefully help others along

[08:34 - 08:38] the way. Your motivation might be totally different, but whatever it is, try to tie

[08:38 - 08:43] this work to a larger goal. In terms of optimizing your study time, I've found that one of the most

[08:43 - 08:47] effective methods is writing notes in my own words. Don't just copy definitions or formulas.

[08:47 - 08:51] Take time to summarize concepts as if you were explaining them to someone else, or really to

[08:51 - 08:55] future you. For example, if you're learning about derivatives, you might write,

[08:55 - 09:00] a derivative measures how a function changes given a tiny nudge to its input. This forces

[09:00 - 09:04] you to actively process the material. Relatedly, when it comes to math formulas, don't just stare

[09:04 - 09:08] at them. I know it's easy to do. It's really helpful to translate them into plain English or

[09:08 - 09:13] whatever spoken language you prefer. For instance, take the equation y equals mx plus b. You might

[09:13 - 09:18] describe m as the slope that shows how steep the line is and b as the point where the line crosses

[09:18 - 09:24] the y-axis. So the final formula might be the value of y, the output, is determined by taking the slope

[09:25 - 09:30] by x, the input, and then adding b, the starting point where the line intersects the y-axis.

[09:30 - 09:33] You can even use your notes as like a personal blog, kind of.

[09:33 - 09:37] Writing short posts about what you've learned is a really solid way to clarify your understanding,

[09:38 - 09:42] and teaching others, even if nobody reads it, solidifies the material in your own mind.

[09:42 - 09:47] Plus, sharing your posts on Medium or LinkedIn not only potentially helps others,

[09:47 - 09:50] but also allows you to build kind of a portfolio showcasing your learning journey.

[09:51 - 09:54] Also, trust me, when it's interview time, you're going to be really happy that you have these

[09:54 - 09:55] notes. I use my own study time, so I'm going to use my own study time. I'm going to use my own

[09:55 - 09:59] study notes all the time. The next piece of advice I have might not be super duper fun,

[09:59 - 10:04] but I also recommend not using just one resource. Personally, I've had a lot of success from taking

[10:04 - 10:08] many different courses and kind of just throwing all my notes together at first. Then I'll write

[10:08 - 10:12] a blog, like I was just talking about, that summarizes all of my learnings. There are a

[10:12 - 10:16] couple of advantages to this approach. First, repetition helps you retain things. If I see

[10:16 - 10:21] a concept multiple times, explained from multiple angles, I'm much more likely to actually get

[10:21 - 10:25] what's going on and remember it for longer than a day. Plus, not only do I see the information

[10:25 - 10:29] presented to me multiple times, I'm writing the concepts out in my own words multiple times,

[10:30 - 10:33] including that final time where I synthesize it all and get ready to share with others.

[10:33 - 10:37] So I have to be really confident that I actually got it by the end if I'm going to be putting it

[10:37 - 10:41] out publicly. Let me show you an example. Here are three different explanations for the concept

[10:41 - 10:46] of rank in linear algebra. Starting with three blue, one brown, you can see this visual intuition,

[10:46 - 10:50] which is a great starting point, but you know, it lacks a little detail. Then we can look at how

[10:50 - 10:55] rank was presented in deeplearning.ai, which has some more detail, but is still presented in a

[10:55 - 10:59] really simple way. Finally, we can go to the Mathematics for Machine Learning ebook to get

[10:59 - 11:03] an even more formalized definition with additional detail. Each one of these builds on the other and

[11:03 - 11:07] adds to my store of knowledge. Finally, once you've built that foundation and get to the level

[11:07 - 11:11] of math where you can actually use it for stuff, I really recommend coding concepts from scratch.

[11:11 - 11:15] If you can code gradient descent or logistic regression using just NumPy, you're off to a

[11:15 - 11:19] really strong start. While I know at this point you are super excited to start learning math,

[11:19 - 11:23] I do want to just circle back to the important fact that if you're a beginner trying to get your

[11:23 - 11:28] first job, in my opinion math should not be the first thing you prioritize. It's really unlikely

[11:28 - 11:32] that your math skills are going to be the thing that gets you a job as a data scientist or machine

[11:32 - 11:37] learning engineer. Instead, prioritize gaining hands-on experience by working on projects and

[11:37 - 11:41] actually building stuff. Employers are far more interested in seeing what you can do with the

[11:41 - 11:45] tools and knowledge you already have than how many formulas you've memorized. As you encounter

[11:45 - 11:48] challenges in your work, you'll naturally be motivated to learn the math behind the algorithms.

[11:49 - 11:53] Remember, math is a tool to help you succeed and shouldn't be a barrier to getting started.

Now,

[11:53 - 11:55] if you're curious about what else you need to know to become a data scientist,

[11:55 - 12:00] I have a really comprehensive video that lays out the entire roadmap from absolute beginner

[12:00 - 12:04] to your first job, including skills to learn, project examples, how to get experience,

[12:04 - 12:08] preparing for interviews, and more. Definitely check that out next!

==========

TRANSLATION (ID):

Apakah Anda ingin menjadi insinyur ilmuwan data (data scientist) atau pembelajaran mesin (machine learning), tetapi Anda merasa terintimidasi oleh semua orang matematika yang terlibat? Saya mengerti, saya pernah ke sana. Saya putus sekolah setelah kelas 10, jadi saya tidak pernah belajar matematika selain trigonometri di sekolah. Saat saya memulai perjalanan saya ke mesin belajar, saya bahkan tidak tahu apa itu turunan. Maju cepat ke hari ini, dan saya sudah melamar ilmuwan di Amazon, dan saya merasa cukup percaya diri dengan kemampuan matematika saya. Aku sudah mengambil yang diperlukan matematika sepanjang jalan menggunakan sumber daya gratis dan pembelajaran mandiri. Hari ini aku akan berjalan Anda melalui beberapa buku, kursus, dan saluran YouTube favorit saya yang membantu saya mencapai tujuan Saya hari ini, dan saya juga akan berbagi beberapa tips belajar tentang cara belajar yang efektif dan tidak menyia-nyiakan waktu Anda waktu berjuang dan merasa bosan. Jadi mari kita mulai. Pertama, mari kita jawab pertanyaan umum. Apakah kamu bahkan benar-benar perlu mengetahui matematika untuk bekerja di pembelajaran mesin (machine learning)? Jawaban singkatnya adalah, itu tergantung pada apa yang ingin Anda lakukan. Untuk peran yang banyak melakukan penelitian saat Anda membuat pembelajaran mesin (machine learning) baru algoritma (algorithm)s, maka ya, Anda jelas perlu mengetahui matematika. Tetapi jika Anda bertanya pada diri sendiri apakah Anda perlu belajar matematika, kemungkinan besar itu bukan pekerjaan yang Anda cari. Bagi praktisi, kebanyakan dari kita di industri (industry), Anda bisa menjadi benar-benar kompeten tanpa mengetahui semua hal yang mendasarinya detail. Pada titik ini, perpustakaan seperti NumPy, Scikit Learn, dan tensorflow (tensorflow) menangani sebagian besar tugas berat mengangkat untukmu. Anda tidak perlu mengetahui matematika di balik penurunan gradien (gradient descent) untuk menerapkan model (model) ke produksi. Jika anda seorang pemula yang mencoba masuk ke pembelajaran mesin (machine learning), menurut saya tidak strategis untuk menghabiskan banyak waktu menghafal rumus atau mempelajari aljabar linier. Anda harus menghabiskan waktu itu untuk membangun sesuatu. Latih model (model) sederhana, jelajahi data Anda, buat pipa (pipeline) yang memprediksi sesuatu yang menyenangkan. Meskipun demikian, ada saat-saat di mana mengetahui matematika sangat membantu. Berikut beberapa

contohnya. Jadi bayangkan Anda pelatihan (training) dan model (model), dan itu tidak menyatu. Jika Anda memahami konsep seperti gradien dan fungsi pengoptimalan, Anda akan mengetahui apakah Anda perlu memahaminya sesuaikan laju pembelajaran (learning rate) Anda, coba pengoptimal lain, atau sesuaikan proses (process) data Anda. Atau katakanlah Anda menjalankan regresi linier (linear regression), dan Anda menafsirkan koefisiennya. Tanpa pengetahuan matematika, Anda mungkin melewatkan soal seperti multikolinearitas, yang menghasilkan koefisien tidak dapat diandalkan. Kemudian Anda mungkin membuat kesimpulan yang salah dari data tersebut, dan merugikan perusahaan jutaan dan kehilangan pekerjaan Anda. Cuma bercanda. Agak. Kita memang harus berhati-hati saat membuatnya keputusan bisnis dari model (model)s yang kami bangun. Jadi selagi Anda bisa, dan harus, memulai tanpa pengetahuan matematika yang mendalam, Anda harus berhati-hati saat mengambil keputusan bisnis dari model (model)s yang kita miliki membangun. Sangat masuk akal untuk membangun kenyamanan Anda dengan matematika seiring berjalannya waktu. Setelah Anda siap, Anda akan mulai menghadapi masalah yang secara alami mendorong Anda untuk belajar lebih banyak. Saat Anda perlu debug (debug) atau jelaskan hasilmu, saat itulah matematika akan mulai diklik karena terhubung untuk sesuatu yang nyata. Serius, jangan biarkan rasa takut terhadap matematika menghentikan Anda untuk memulai. Anda tidak perlu melakukannya pelajari semuanya terlebih dahulu untuk membuat kemajuan. Jadikan tangan Anda kotor dengan alat (tool)s, buat porta (port)folio Anda, dan biarkan matematika berkembang sebagai keterampilan seiring dengan pengetahuan praktis Anda. Baiklah, sekarang mari kita bicara tentang apa yang harus dipelajari saat Anda membangun landasan matematika untuk pekerjaan pembelajaran mesin (machine learning). Pertama, aljabar linier. Aljabar linier merupakan dasar untuk pembelajaran mesin (machine learning), terutama untuk pembelajaran mendalam (deep learning). Banyak model (model)s mengandalkan representasi data dan komputasi sebagai matriks dan vektor (vector)s. Inilah yang harus diprioritaskan. Pertama, matriks dan vektor (vector)s. Bayangkan matriks sebagai kisi-kisi angka dan vektor (vector)s sebagai daftar angka. Data sering kali disimpan dengan cara ini, dan operasi seperti penjumlahan, perkalian, dan perkalian titik sangat penting dalam cara model (model)s proses (process) informasi tersebut. Penentu dan invers. Penentu memberi tahu Anda apakah suatu matriks dapat dibalik, yang digunakan dalam masalah optimasi dan menyelesaikan sistem (system)s persamaan linier. Nilai eigen dan eigenvektor (vector)s. Ini adalah kunci untuk memahami varians (variance) dalam data dan merupakan dasar dari teknologi (tech)niques seperti analisis komponen utama (principal component analysis), yang dapat membantu mengurangi dimensi kumpulan data. Terakhir, dekomposisi matriks. Metode seperti dekomposisi nilai tunggal digunakan dalam rekomendasi sistem (system)s, reduksi dimensi, dan data mampatkan (compress)ion. Sekarang kita masuk ke kalkulus dasar. Kalkulus adalah inti untuk memahami cara model (model)s belajar dari data. Namun kita tidak perlu khawatir untuk memecahkan masalah yang super rumit. Hanya saja tentang menangkap beberapa ide kunci. Pertama, turunan dan gradien. Derivatif mengukur bagaimana sesuatu berubah, dan gradien, yang merupakan turunan multidimensi, kita sebut gradien. pengoptimalan daya algoritma (algorithm)s seperti penurunan gradien (gradient descent). Ini membantu model (model)s menyesuaikan parameter (parameter)s mereka untuk meminimalkan galat (error). Aturan rantai sangat penting dalam jaringan saraf (neural network)s. Begitulah cara kerja propagasi balik (backpropagation), yang merupakan proses (process) untuk

mencari tahu seberapa besar kontribusi masing-masing bobot (weight) di jaringan (network) terhadap overall galat (error) so the model (model) can learn effectively. Terakhir, dasar-dasar pengoptimalan. Konsepnya seperti lokal versus minimum global, titik pelana, dan konveksitas sangat penting untuk memahami mengapa beberapa model (model) mendapatkan terjebak dan yang lain berhasil menemukan solusi terbaik. Terakhir, statistik dan probabilitas (probability). Statistik dan probabilitas (probability) adalah inti dari pemahaman data. Selagi mereka lebih terkait dengan sains data (data science), pasti ada banyak nilai untuk pembelajaran mesin (machine learning) sebagai Sehat. Inilah yang perlu Anda ketahui. Distribusi. Kenali distribusi umum seperti normal, binomial, dan seragam. Distribusi normal khususnya muncul dimana-mana di sains data (data science) dan pembelajaran mesin (machine learning). varians (variance) dan bersamavarians (variance). varians (variance) memberitahu Anda bagaimana penyebarannya data Anda, sementara covarians (variance) menunjukkan hubungan dua variabel (variable). Konsep-konsep ini sebenarnya importa (port)ant untuk seleksi fitur (feature selection) dan memahami struktur data Anda. Teorema Bayes. Meskipun sudah agak konyol, mungkin nama yang mengintimidasi, teorema Bayes adalah alat (tool) yang cukup sederhana namun kuat untuk penalaran probabilistik. Ini dasar untuk algoritma (algorithm)s seperti naive bayes (naive bayes), kejutan besar, yang digunakan untuk hal-hal seperti deteksi spam (spam), serta untuk optimasi bayesian (bayesian) untuk hyperparameter (hyperparameter) penyetelan. Anda juga ingin memahami estimasi kemungkinan maksimum, yang membantu memperkirakan model (model) parameter (parameter)s dengan menemukan nilai yang memaksimalkan kemungkinan data Anda. Itu sungguh konsep dasar, terutama di algoritma (algorithm)s seperti regresi logistik (logistic regression). Terakhir, pengambilan sampel dan bersyarat probabilitas (probability). Pengambilan sampel memungkinkan Anda bekerja dengan subkumpulan data secara efisien dan bersyarat probabilitas (probability) penting untuk memahami hubungan antar peristiwa, terutama di bayesian (bayesian) model (model)s. Ini jelas bukan daftar yang lengkap, tapi menurut saya ini adalah tampilan (view) konsep umum yang bagus Anda harus tahu untuk melakukan pekerjaan dengan baik sebagai ilmuwan data (data scientist) yang fokus pada pembelajaran mesin (machine learning) atau sebagai seorang insinyur pembelajaran mesin (machine learning). Selanjutnya, saya akan membagikan sumber daya terbaik untuk mempelajari konsep-konsep ini tanpa itu membuat stres atau membebani, dan semoga juga tidak terlalu membosankan. Secara pribadi, saya akan sangat menyukainya merekomendasikan memulai dengan visual (visual) dan pemahaman intuitif tentang konsep-konsep utama sebelum Anda memulai membaca buku teks (text) yang sangat sulit dan mencoba menyelesaikan persamaan. Untuk aljabar linier dan kalkulus, Saya tidak dapat berbicara cukup banyak tentang seri Essence of Linear Algebra dan Essence of Calculus 3Blue1Brown. video (video) ini memberikan pengenalan yang solid tentang apa yang sebenarnya diukur dan dimanipulasi ketika kita menggunakan pendekatan matematika ini. Lebih menariknya lagi, mereka menunjukkan keindahan di dalamnya. Agak aneh untuk mengatakan bahwa matematika video (video)s bisa menjadi inspirasi, tetapi yang ini adalah inspirasinya. Untuk statistik dan probabilitas (probability), saya penggemar berat StatQuest. video (video)nya jelas, menarik, dan menyenangkan untuk ditonton. StatQuest memiliki daftar putar dengan lebih daritampilan (view)s pada statistik inti dan konsep pembelajaran mesin (machine learning), jadi saya sarankan memulai dari sana. Setelah Anda memiliki intuisi visual (visual), Anda dapat mulai mempelajari buku dan kursus yang

lebih terstruktur. Ada banyak hal hebat pilihan di sini juga. Mari kita lihat beberapa yang saya pribadi gunakan untuk belajar. Saya menyelesaikan Matematika untuk spesialisasi pembelajaran mesin (machine learning) dari Imperial College London di Coursera ketika I baru saja dimulai. Peminatan ini dibagi menjadi tiga mata kuliah, aljabar linier, kalkulus multivariat, dan yang terakhir di analisis komponen utama (principal component analysis). Kursus-kursusnya terstruktur dengan baik dan menyertakan campuran video (video) ceramah, kuis, dan program (program) tugas dengan Python. Saya menemukan Tentu saja menjadi sedikit menantang sebagai pemula, tapi itu adalah overtampilan (view) yang sangat bagus dan saya berhasil melewati semuanya dengan sedikit usaha. DeepLearning.ai juga baru-baru ini merilis a spesialisasi di Coursera. Spesialisasi ini memiliki kursus tentang aljabar linier dan kalkulus, tapi alih-alih PCA, kursus terakhir berfokus pada statistik dan probabilitas (probability). Saya pribadi sedang bekerja melalui spesialisasi ini saat ini dan secara keseluruhan menurut saya ini adalah pilihan yang sangat bagus. Setiap modul dimulai dengan motivasi yang bagus tentang bagaimana matematika terhubung dengan pembelajaran mesin (machine learning) yang diterapkan konsep. Ini memiliki latihan pengkodean dengan Python dan beberapa alat (tool) 3D yang rapi untuk dipusingkan untuk mendapatkan pemahaman visual (visual) yang baik tentang konsep juga. Jika Anda lebih suka belajar dari buku, Saya punya beberapa saran di sana juga. Pertama, jika Anda menyukai anime atau hal-hal kutu buku, ya ampun, saya punya rekomendasi untuk Anda. Tahukah Anda bahwa mereka memiliki buku matematika manga? Jadi saya sudah membaca Manganya panduan (guide) ke Kalkulus, Manga panduan (guide) ke Aljabar Linier, dan Manga panduan (guide) ke Statistik. Mereka juga memiliki ton lebih. Menurut saya, mereka memiliki beberapa hal seperti fisika, dan beberapa konsep teknik lainnya. Jadi memang begitu sangat sangat menyenangkan. Saya tidak bisa mengatakan kualitas pengajarannya seperti kelas dunia atau apa pun, tapi mereka lucu dan menarik dan membuatku tidak takut membaca buku matematika. Level berikutnya adalah buku matematika sungguhan. Ini adalah beberapa yang terbaik. Matematika untuk pembelajaran mesin (machine learning) buku elektronik (ebook) oleh Dysonroth dan rekannya sangat bagus, sumber daya yang komprehensif dan tersedia gratis untuk penggunaan pribadi. Ini mencakup topik-topik utama kita sudah membahas seperti aljabar linier, kalkulus, probabilitas (probability), dan optimasi, dengan fokus pada bagaimana konsep ini diterapkan pada pembelajaran mesin (machine learning) algoritma (algorithm)s. Itu relatif ramah bagi pemula dan umumnya dianggap sebagai salah satu buku terbaik untuk mempelajari materi ini. Berikutnya, Statistik Praktis untuk ilmuwan data (data scientist)s adalah sumber daya populer lainnya yang menyertakan kode contoh dalam Python dan R. Sekarang sebelum kita benar-benar mulai belajar, saya pikir sangat penting untuk menghabiskan waktu sedikit waktu untuk memikirkan secara mendalam mengapa Anda ingin melakukan ini. Secara pribadi, saya menemukannya belajar hanya karena saya merasa harus melakukannya atau karena itu tugas yang sewenang-wenang. saya mengerti mudah teralihkan dan saya tidak benar-benar menyimpan banyak informasi. Sebaliknya, saya mencoba terhubung dengan motivasi yang lebih dalam, jika memungkinkan. Secara pribadi, saat ini saya memiliki motivasi yang sangat mendasar. Saya ingin melakukan pekerjaan saya dengan baik karena Saya ingin mendapatkan banyak uang sehingga saya bisa menjaga semua orang yang saya cintai. Saya punya ini bagus sumber daya untuk dipelajari yang bahkan tidak tersedia beberapa tahun yang lalu, dan saya memiliki opporta (port)unity ini untuk mendorong diri

sendiri dan memastikan bahwa semua orang yang saya cintai aman dan terlindungi saat ini dan di masa depan. Jadi ini bukan untuk memberi tekanan ekstra pada diri saya sendiri, tetapi sebenarnya hanya cara yang berhasil bagi saya untuk bersikap baik Saya merasa senang karena saya memiliki kesempatanporta (port)kesatuan ini untuk belajar dan berkembang dan semoga dapat membantu orang lain jalannya. Motivasi Anda mungkin sangat berbeda, tetapi apa pun itu, cobalah untuk mengikatnya ini bekerja untuk tujuan yang lebih besar. Dalam hal mengoptimalkan waktu belajar Anda, saya menemukan itu salah satu yang paling banyak metode yang efektif adalah menulis catatan dengan kata-kata saya sendiri. Jangan hanya menyalin definisi atau rumus. Luangkan waktu untuk meringkas konsep seolah-olah Anda sedang menjelaskannya kepada orang lain, atau sebenarnya kepada orang lain masa depan kamu. Misalnya, jika Anda mempelajari turunan, Anda bisa menulis, turunan mengukur bagaimana suatu fungsi berubah dengan sedikit dorongan pada masukannya. Ini memaksa Anda untuk secara aktif proses (process) materi. Terkait dengan rumus matematika, jangan hanya menatap saja pada mereka. Saya tahu itu mudah dilakukan. Sangat membantu untuk menerjemahkannya ke dalam bahasa Inggris atau bahasa lisan apa pun yang Anda sukai. Untuk instans (instance), ambil persamaan y sama dengan mx ditambah b. Anda mungkin gambarkan m sebagai kemiringan yang menunjukkan kecuraman garis dan b sebagai titik perpotongan garis sumbu y. Jadi rumus akhirnya mungkin nilai y, keluaran (output), ditentukan dengan mengambil kemiringan dengan x, masukan, lalu tambahkan b, titik awal perpotongan garis dengan sumbu y. Anda bahkan dapat menggunakan catatan Anda seperti bcatatan (log) pribadi. Menulis postingan singkat tentang apa yang telah Anda pelajari adalah cara yang sangat tepat untuk memperjelas pemahaman Anda, dan mengajar orang lain, bahkan jika tidak ada yang membacanya, akan memperkuat materi dalam pikiran Anda. Selain itu, membagikan postingan Anda di Medium atau tautan (link)edIn tidak hanya berpotensi membantu orang lain, tetapi juga memungkinkan Anda membuat semacam porta (port)folio yang menampilkan perjalanan belajar Anda. Juga, percayalah, ketika waktunya intertampilan (view), Anda akan sangat senang memiliki ini catatan. Saya menggunakan waktu belajar saya sendiri, jadi saya akan menggunakan waktu belajar saya sendiri. Aku akan menggunakan milikku sendiri catatan belajar sepanjang waktu. Nasihat saya berikutnya mungkin tidak terlalu menyenangkan, tapi saya juga menyarankan untuk tidak menggunakan satu sumber saja. Secara pribadi, saya telah memperoleh banyak keberhasilan dalam pengambilan banyak kursus berbeda dan hanya menyatukan semua catatan saya pada awalnya. Lalu aku akan menulis a bcatatan (log), seperti yang baru saja saya bicarakan, yang merangkum semua pembelajaran saya. Ada sebuah beberapa advantag (tag)es untuk pendekatan ini. Pertama, pengulangan membantu Anda mempertahankan sesuatu. Jika saya melihat sebuah konsep berkali-kali, dijelaskan dari berbagai sudut, kemungkinan besar saya akan benar-benar memahaminya apa yang terjadi dan mengulanginya selama lebih dari satu hari. Ditambah lagi, saya tidak hanya melihat informasinya disajikan kepada saya berkali-kali, saya menuliskan konsepnya dengan kata-kata saya sendiri berkali-kali, termasuk saat terakhir saya menyatukan semuanya dan bersiap untuk berbagi dengan orang lain. Jadi saya harus benar-benar yakin bahwa saya benar-benar mendapatkannya pada akhirnya jika saya ingin memasangnya keluar secara publik. Mari saya tunjukkan sebuah contoh. Berikut tiga penjelasan berbeda untuk konsep tersebut peringkat dalam aljabar linier. Dimulai dengan tiga warna biru, satu coklat, Anda dapat melihat intuisi visual (visual) ini, yang merupakan titik awal yang bagus,

tapi tahukah Anda, ada sedikit detail yang kurang. Lalu kita bisa melihat caranya peringkat disajikan di deeplearning.ai, yang memiliki beberapa detail lebih lanjut, namun tetap disajikan dalam a cara yang sangat sederhana. Terakhir, kita bisa pergi ke Matematika untuk pembelajaran mesin (machine learning) buku elektronik (ebook) untuk mendapatkan definisi yang lebih formal dengan detail tambahan. Masing-masing dibangun di atas yang lain dan menambah simpanan ilmu saya. Terakhir, setelah Anda membangun fondasi itu dan mencapai level tersebut matematika di mana Anda benar-benar dapat menggunakannya untuk berbagai hal, saya sangat merekomendasikan pengkodean konsep dari awal. Jika Anda dapat membuat kode penurunan gradien (gradient descent) atau regresi logistik (logistic regression) hanya dengan menggunakan NumPy, Anda akan menuju ke awal yang sangat kuat. Meskipun saya tahu saat ini Anda sangat bersemangat untuk mulai belajar matematika, Saya hanya ingin mengingat kembali fakta penting bahwa jika Anda seorang pemula mencoba untuk mendapatkan pekerjaan pertama, menurut saya matematika tidak boleh menjadi hal pertama yang diprioritaskan. Ini sangat tidak mungkin bahwa kemampuan matematika Anda akan menjadi hal yang memberi Anda pekerjaan sebagai ilmuwan data (data scientist) atau mesin insinyur pembelajaran. Sebaliknya, prioritaskan untuk mendapatkan pengalaman langsung dengan mengerjakan proyek (project)s dan sebenarnya membangun sesuatu. Pengusaha jauh lebih tertarik untuk melihat apa yang dapat Anda lakukan dengan perusahaan tersebut alat (tool)s dan pengetahuan yang sudah Anda miliki dibandingkan berapa banyak rumus yang sudah Anda hafal. Seperti yang Anda temui tantangan dalam pekerjaan Anda, secara alami Anda akan termotivasi untuk mempelajari matematika di balik algoritma (algorithm)s. Reanggota (member), matematika adalah alat (tool) untuk membantu Anda sukses dan tidak boleh menjadi penghalang untuk memulai. Sekarang, jika Anda penasaran tentang apa lagi yang perlu Anda ketahui untuk menjadi ilmuwan data (data scientist), Saya memiliki video (video) yang sangat komprehensif yang memaparkan keseluruhan peta jalan dari pemula ke pekerjaan pertama Anda, termasuk keterampilan untuk dipelajari, proyek (project) contoh, cara mendapatkan pengalaman, mempersiapkan intertampilan (view)s, dan banyak lagi. Pastikan untuk memeriksanya selanjutnya!